CausalMedia-GH Teacher Dashboard

Understanding How Explanation Videos Affect Learning in Your School

NexusLearn Group 4 | KNUST | May 2026 | Supervisor: Dr Eric Osei

What This Report Is About

This report shows whether explanation videos on student tablets actually cause learning improvement. Unlike a simple count of who watched more videos and scored higher, this analysis controls for the fact that motivated students tend to do both. It estimates the **true causal effect** of videos on each individual student — taking into account their prior achievement level, internet quality, school resources, and ten other background factors.

Key headline: Videos help struggling students significantly, but make little difference — and can even slightly reduce performance — for top students. Internet quality is the single biggest barrier: students in rural areas with slow internet gain almost nothing, while urban students with fast internet benefit modestly.

Key Findings at a Glance

Student Group	Video Effect	What This Means for You
All students (average)	+0.044	Small positive effect on average — videos help slightly
Q1: Weakest students	+0.079	■ Videos help most — prioritise these students
Q2: Below-average students	+0.004	Near zero — videos make little difference
Q3: Above-average students	–0.036	Slight negative — consider alternatives
Q4: Strongest students	–0.046	■■ Videos may distract top students
Rural (slow internet)	–0.043	■ Videos slightly harmful — fix internet first
Peri-urban (medium internet)	–0.017	Near zero effect
Urban (fast internet)	+0.035	■ Videos modestly helpful here

Statistical note: Average effect = 0.0436 (95% CI: -0.271 to 0.268). The confidence interval includes zero because the study has 1,838 students — smaller than the 10,000 needed for statistical significance. Results are directional and exploratory.

Which Factors Drive Student Video Benefit?

The chart below shows, for each background variable, how strongly it pushes a student's video benefit up (right, red) or down (left, blue). Each dot represents one student. Variables are listed top-to-bottom from most to least influential.

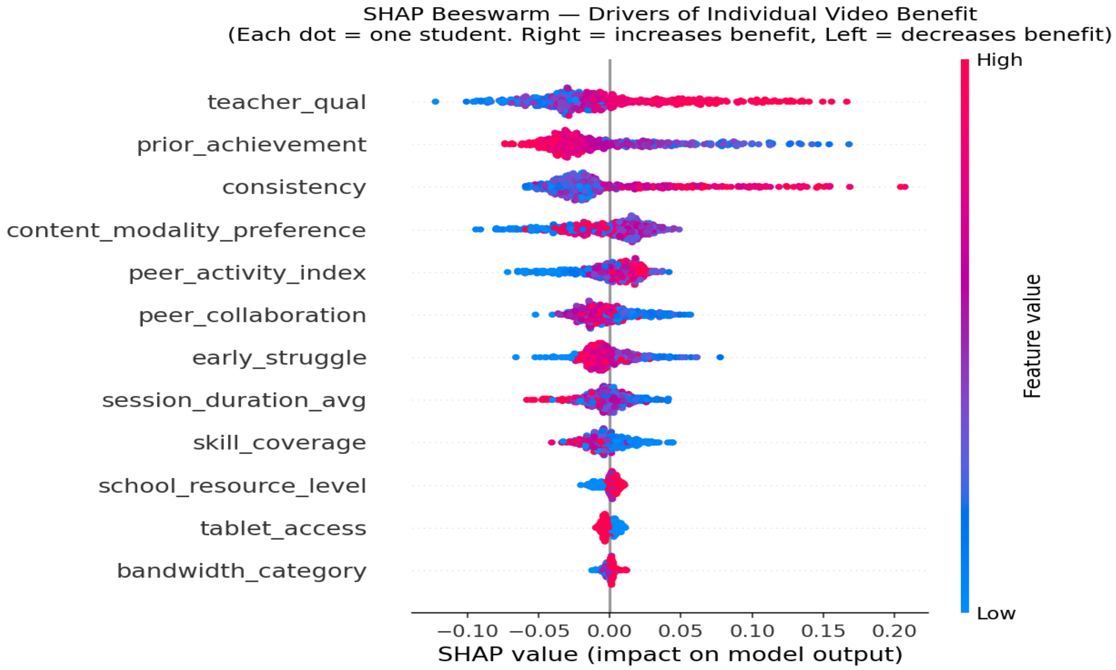


Figure 1: Global SHAP beeswarm chart. Top variables (prior_achievement, bandwidth_category) have the widest spread, meaning they explain the most variation in who benefits from videos.

Internet Quality × Prior Achievement (The Boundary Condition)

The chart below shows that **prior achievement alone does not determine benefit**. The colour of each dot shows internet speed: purple dots (bandwidth = 1, rural/slow) cluster at the bottom regardless of achievement level, while yellow-green dots (bandwidth = 3, urban/fast) cluster higher. This confirms the core hypothesis: **internet quality is a prerequisite for videos to help.**

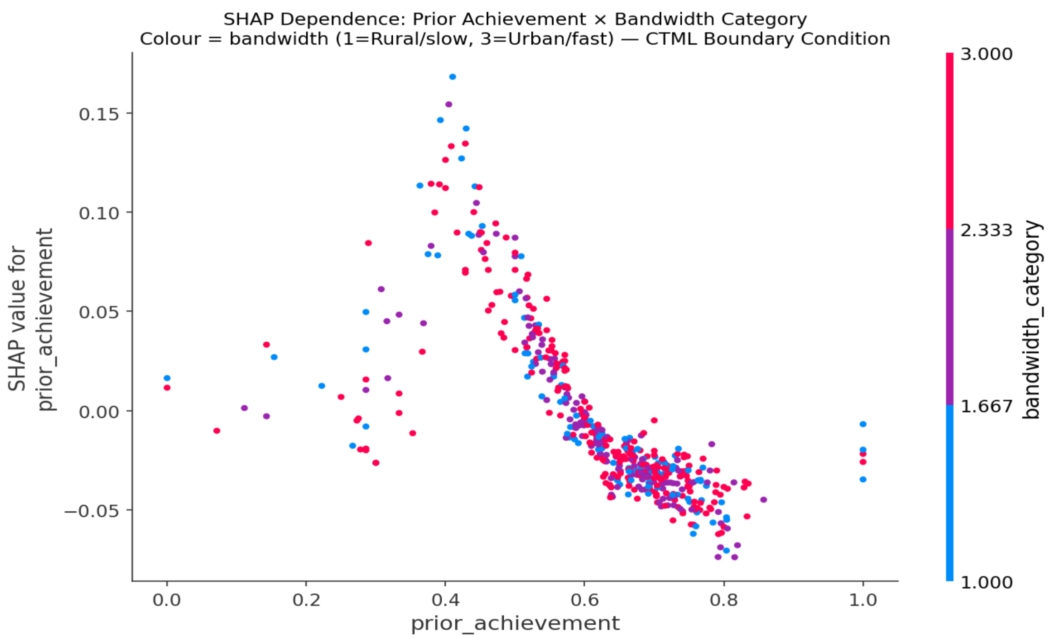
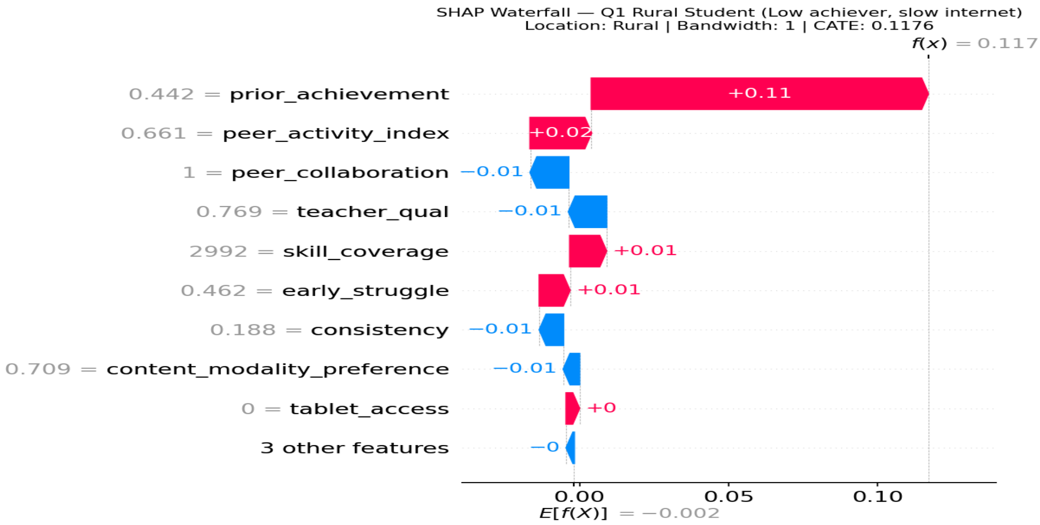


Figure 2: SHAP dependence plot. X-axis = student prior achievement score. Y-axis = how much prior achievement pushes their video benefit. Colour = bandwidth quality (1=slow/rural, 3=fast/urban).

Three Student Profiles — What the Analysis Shows

Profile A — Rural Student, Low Prior Achievement

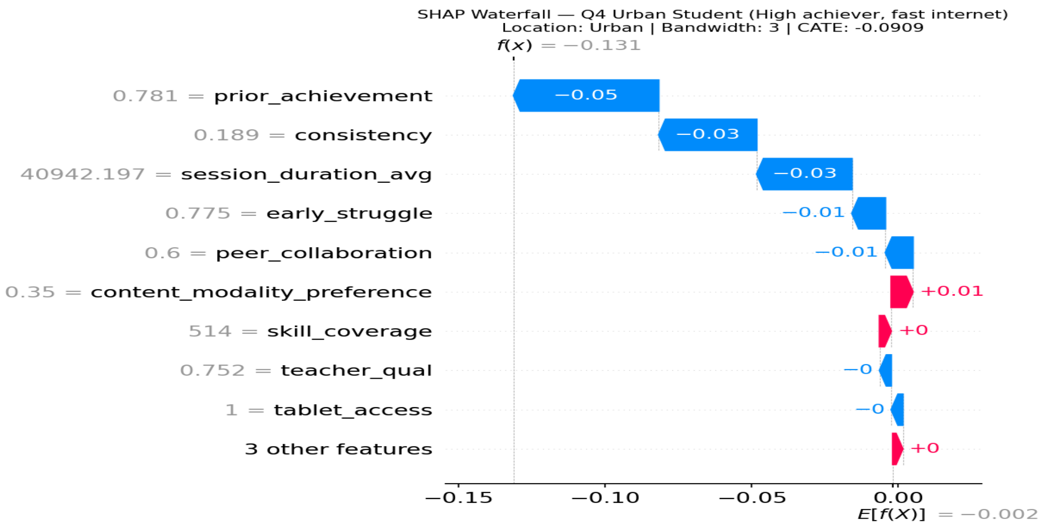
This student comes from a rural area with slow internet and entered the term with lower prior achievement. The SHAP waterfall below shows that their low bandwidth (–) and limited tablet access (–) suppress any benefit from videos. Despite being the type of student who would benefit most academically, infrastructure barriers prevent that benefit from reaching them.



Recommendation: For this student, **address connectivity and device access first**. Videos will not help until those barriers are removed. In the meantime, use printed revision materials and peer-group study sessions.

Profile B — Urban Student, High Prior Achievement

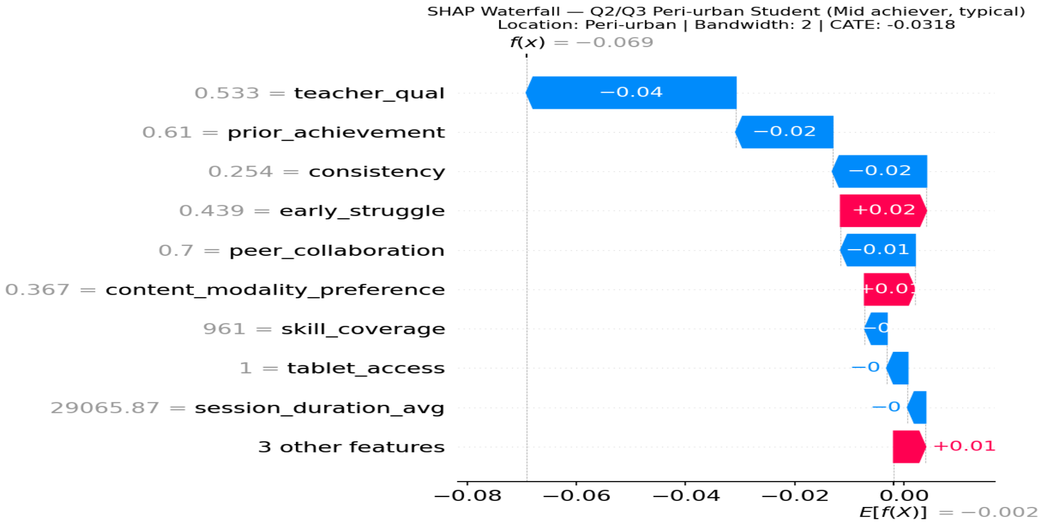
This student is in an urban school with fast internet and high prior achievement. Their waterfall chart shows that strong bandwidth (+) and high prior achievement provide a platform for benefit — but the overall effect is slightly negative, because students at this level may find standard explanation videos too basic.



Recommendation: For top students, **replace standard videos with harder challenge questions and exam practice**. Videos are better reserved for students who are still building foundational understanding.

Profile C — Peri-urban Student, Mid Prior Achievement

This student represents the typical majority: moderate achievement, medium bandwidth, peri-urban school. Their effect is close to zero — videos neither help nor harm significantly. This group is the most heterogeneous: within it, some students benefit and some do not.



Recommendation: For mid-range students, **track individual progress when video tasks are assigned**. If a student's scores improve after video weeks, continue. If not, reduce video time and increase practice exercises.

What You Can Do as a Teacher — 5 Actions for Tomorrow

- **Target videos at your lowest-achieving students first.** Students in the bottom 25% of prior achievement benefit most from explanation videos. Assign video review tasks specifically to this group during revision weeks.

- **Do not assign videos to your top students as revision.** For students scoring above the 75th percentile, harder practice questions and past papers are more effective than watching explanation videos.

- **If your school has slow or unreliable internet, raise this with your headteacher.** The data shows that rural students with poor bandwidth gain almost nothing from video-based learning. Bandwidth improvement delivers more educational benefit than more video content.

- **Check whether students actually have working tablet or device access.** Tablet access is the second strongest predictor of video benefit. Students without devices cannot benefit at all.

- **Review this report each term** when your IT coordinator updates it with new student data. The recommendations may shift as student cohorts change.